

Application Security Foundations

CST-407

4 Credits

Sep 2 - Dec 14

Course Description

This course examines security principles for application developers. The course also examines common security vulnerabilities found in modern dynamic web applications, secure programming practices, and how to avoid and illuminate the common security vulnerabilities. Prerequisite: CST-339.

Instructor Contact Information

Shad Sluiter

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Class Resources

CST-407 Milestone Assignments (REQUIRED)

Refer to this resource when completing the milestone assignments in this course.

<https://mygcuedu6961.sharepoint.com/:w:/r/sites/CSETGuides/CST/CST-407/CST-407-RS-MilestoneAssignments.docx?d=wcf77e79906e84b45947e30adc59fa374&csf=1&web=1&e=A7jElm>

Visual Studio Code (REQUIRED)

This course requires the use of Visual Studio Code.

Be sure to download or update to the version specified by your instructor.

https://www.gcumedia.com/digital-resources/microsoft/2019/visual-studio-code_1e.php

GPG (REQUIRED)

This course requires the use of GPG.

Be sure to download or update to the version specified by your instructor.

http://www.gcumedia.com/digital-resources/gpg-tools/2018/gpg-suite_1e.php

MAMP (REQUIRED)

This course requires the use of MAMP.

Be sure to download or update to the version specified by your instructor.

Refer to the "CST-407 MAMP Installation Instructions," when downloading this software.

https://www.gcumedia.com/digital-resources/mamp-and-mamp-pro/2014/mamp_3e.php

CST-407 MAMP Installation Instructions (REQUIRED)

Refer to this resource when downloading MAMP.

CST-407-RS-MAMPInstallationInstructions.docx

Wireshark (REQUIRED)

This course requires the use of Wireshark.

Be sure to download or update to the version specified by your instructor.

https://www.gcumedia.com/digital-resources/wireshark-foundation/2016/wireshark_2e.php

Topic 1: The ABCs of Information Security

Sep 2, 2025 - Sep 7, 2025 Max Points: 45

Information security professionals play an important role in ensuring that confidentiality, integrity, and availability (CIA) are maintained at the highest level for any information system. Their focus is on the protection of data (information assets), not just from unauthorized access or use, but from being compromised during critical situations (natural disasters, computer/server malfunctions, etc.). In order to provide this protection, they must have a solid foundation in risk management and information security best practices.

Objectives:

1. Identify compromised aspects of the CIA triad and evaluate detection mechanisms in real-world data breaches.
2. Compare the effectiveness of different detection and mitigation strategies in cybersecurity.
3. Identify attacker methods in past security incidents and propose preventive measures.
4. Recall key concepts related to the causes, impacts, and mitigation strategies of high-profile data breaches.

Resources

Activity 1: The ABCs of Information Security (REQUIRED)

Refer to this resource when completing "Activity 1: The ABCs of Information Security."

<https://mygcu.edu6961.sharepoint.com/:w:/r/sites/CSETGuides/CST/CST-407/CST-407-RS-T1-Activity-1-The-ABCs-of-Information-Security.docx?d=w0bdf91a16a544147a5fccd2f207533b4&csf=1&web=1&e=lb70Vh>

Assessments

Summary of Current Course Content Knowledge

Start Date & Time	Due Date & Time	Points
Sep 2, 2025, 12:00 AM	Sep 2, 2025, 11:59 PM	0

Assessment Description

Academic engagement through active participation in instructional activities related to the course objectives is paramount to your success in this course and future courses. Through interaction with your instructor and classmates, you will explore the course material and be provided with the best opportunity for objective and competency mastery. To begin this class, review the course objectives for each Topic, and then answer the following questions as this will help guide your instructor for course instruction.

1. Which weekly objectives do you have prior knowledge of and to what extent?
2. Which weekly objectives do you have no prior knowledge of?
3. What course-related topics would you like to discuss with your instructor and classmates? What questions or concerns do you have about this course?

Class Introductions

Start Date & Time	Due Date & Time	Points
Sep 2, 2025, 12:00 AM	Sep 3, 2025, 11:59 PM	0

Assessment Description

Take a moment to explore your new classroom and introduce yourself to your fellow classmates. What are you excited about learning? What do you think will be most challenging?

Activity 1: The ABCs of Information Security

Start Date & Time	Due Date & Time	Points
Sep 2, 2025, 12:00 AM	Sep 7, 2025, 11:59 PM	35
	 Sep 21, 2025, 11:59 PM	

Assessment Traits

 Accommodated

Assessment Description

Refer to "Activity 1: The ABCs of Information Security," located in the topic Resources, for details regarding successful assignment completion.

While APA style is not required for the body of this assignment, solid academic writing is expected, and documentation of sources should be presented using APA formatting guidelines, which can be found in the APA Style Guide, located in the Student Success Center.

This assignment uses a rubric. Review the rubric prior to beginning the assignment to become familiar with the expectations for successful completion.

You are not required to submit this assignment to LopesWrite.

Week 1 Participation

Start Date & Time	Due Date & Time	Points
Sep 2, 2025, 12:00 AM	Sep 7, 2025, 11:59 PM	10

Topic 2: Encryption Technology and Tools

Sep 8, 2025 - Sep 14, 2025 Max Points: 70

Since ancient times, people have tried to communicate in secure channels. Therefore, the field of cryptology has advanced as rapidly as electronics and digital computing. Encryption and hashing are processes of converting data into a cipher to ensure privacy. As an information security professional, it is imperative to understand basic access control, authentication, hashing, and encryption methods used in securing data in information systems.

Objectives:

- 1. Utilize public and private keys to encrypt and decrypt messages securely.
- 2. Evaluate the impact of quantum computing on current encryption technologies.
- 3. Identify key differences between hashing, symmetric encryption, and asymmetric encryption.

Resources

Activity 2: Data Encryption Tutorial (REQUIRED)

Refer to this resource when completing "Activity 2: Data Encryption Tutorial."

<https://mygcuedu6961.sharepoint.com/:w:/r/sites/CSETGuides/CST/CST-407/CST-407-RS-T2-Activity-2-Data-Encryption-Tutorial.docx?d=wd4b7a75e550d4769b13ebfa11fe8cdc3&csf=1&web=1&e=Y4Rz5M>

GnuPG (REQUIRED)

Refer to the GnuPG website when completing Activity 2.

<https://gnupg.org>

Application Layer (REQUIRED)

Read "Application Layer," from Wikipedia.

https://en.wikipedia.org/wiki/Application_layer

Assessments

Activity 2: Data Encryption Tutorial

Start Date & Time	Due Date & Time	Points
Sep 8, 2025, 12:00 AM	Sep 14, 2025, 11:59 PM	40
	 Sep 28, 2025, 11:59 PM	

Assessment Traits

 Accommodated

Assessment Description

Refer to "Activity 2: Data Encryption Tutorial," located in the topic Resources, for details regarding successful assignment completion.

While APA style is not required for the body of this assignment, solid academic writing is expected, and documentation of sources should be presented using APA formatting guidelines, which can be found in the APA Style Guide, located in the Student Success Center.

This assignment uses a rubric. Review the rubric prior to beginning the assignment to become familiar with the expectations for successful completion.

You are not required to submit this assignment to LopesWrite.

Security Terminology Quiz

Start Date & Time	Due Date & Time	Points	Time Limit
Sep 8, 2025, 12:00 AM	Sep 14, 2025, 11:59 PM	20	20 mins
			 30 mins

Assessment Traits

 Accommodated

 Timed

Assessment Description

Complete the quiz as directed by your instructor.

Week 2 Participation

Start Date & Time	Due Date & Time	Points
Sep 8, 2025, 12:00 AM	Sep 14, 2025, 11:59 PM	10

Topic 3: Password Hashing and Cracking

Sep 15, 2025 - Sep 21, 2025 Max Points: 70

In today's digital world, password protection and data integrity are some of the most critical aspects of system security. The process of password hashing provides a layer of security by ensuring that even if passwords are exposed or stolen, they cannot easily be reversed into their original form. Data collections, such as a document, can be insured against modification by using hashing techniques. This lesson introduces password hashing and explores techniques used for both protecting and cracking passwords.

Objectives:

1. Demonstrate the process of hashing and salting passwords for secure storage and authentication.
2. Contrast hashing and encryption processes, distinguishing their roles in data security.
3. Evaluate the importance of using secure hashing algorithms and the concept of rainbow table attacks.
4. Identify the benefits of password hashing and the role of salting in enhancing security.

Resources

How Does Social Behavior Affect Your Password? (REQUIRED)

Read "How Does Social Behavior Affect Your Password?", " by He et al., from IEEE Network (2021).

<https://ieeexplore-ieee-org.lopes.idm.oclc.org/document/9606834>

Activity 3: Password Hashing and Cracking (REQUIRED)

Refer to this resource when completing "Activity 3: Password Hashing and Cracking."

<https://mygcuedu6961.sharepoint.com/:w:/r/sites/CSETGuides/CST/CST-407/CST-407-RS-T3-Activity-3-Password-Hashing-and-Cracking.docx?d=w05b2d7e46f314a00ac4d2c8595506243&csf=1&web=1&e=ytuTpY>

Assessments

Activity 3: Password Hashing and Cracking

Start Date & Time	Due Date & Time	Points
Sep 15, 2025, 12:00 AM	Sep 21, 2025, 11:59 PM	40
	 Oct 5, 2025, 11:59 PM	

Assessment Traits

 Accommodated

Assessment Description

Refer to "Activity 3: Password Hashing and Cracking," located in the topic Resources, for details regarding successful assignment completion.

Access the file "CST-407-RS-T3.Hashing.zip" as directed by the activity guide.

While APA style is not required for the body of this assignment, solid academic writing is expected, and documentation of sources should be presented using APA formatting guidelines, which can be found in the APA Style Guide, located in the Student Success Center.

This assignment uses a rubric. Review the rubric prior to beginning the assignment to become familiar with the expectations for successful completion.

You are not required to submit this assignment to LopesWrite.

Attachments

CST-407-RS-T3.Hashing.zip

Encryption Quiz

Start Date & Time	Due Date & Time	Points	Time Limit
Sep 15, 2025, 12:00 AM	Sep 21, 2025, 11:59 PM	20	20 mins
			 30 mins

Assessment Traits

 Accommodated

 Timed

Assessment Description

Complete the quiz as directed by your instructor.

Week 3 Participation

Start Date & Time	Due Date & Time	Points
Sep 15, 2025, 12:00 AM	Sep 21, 2025, 11:59 PM	10

Topic 4: Secure Network Communication

Sep 22, 2025 - Sep 28, 2025 Max Points: 70

Application security extends beyond the code we write and the server on which it runs. The way in which an application sends data across a network will determine how secure the data remains. We will demonstrate in this lesson that unencrypted network requests are vulnerable to packet sniffers.

Objectives:

1. Configure an application to use SSL/TLS certificates for secure HTTPS communication.
2. Identify common network protocols and their purposes in secure communication.
3. Demonstrate the capture of unencrypted data being transported on a network.
4. Identify vulnerabilities of using unencrypted channels in network communications.

Resources

Activity 4: Wireshark Network Packet Capture Demonstration (REQUIRED)

Refer to this resource when completing "Activity 4: Wireshark Network Packet Capture Demonstration."

<https://mygcuedu6961.sharepoint.com/:w:/r/sites/CSETGuides/CST/CST-407/CST-407-RS-T4-Activity-4-Wireshark-Network-Packet-Capture-Demonstration.docx?d=w7fcbb460f76e4555a126abbf76ae352b&csf=1&web=1&e=bFAezD>

Assessments

Activity 4: Wireshark Network Packet Capture Demonstration

Start Date & Time	Due Date & Time	Points
Sep 22, 2025, 12:00 AM	Sep 28, 2025, 11:59 PM	40
	 Oct 12, 2025, 11:59 PM	

Assessment Traits

 Accommodated

Assessment Description

Refer to "Activity 4: Wireshark Network Packet Capture Demonstration," located in the topic Resources, for details regarding successful assignment completion.

While APA style is not required for the body of this assignment, solid academic writing is expected, and

documentation of sources should be presented using APA formatting guidelines, which can be found in the APA Style Guide, located in the Student Success Center.

This assignment uses a rubric. Review the rubric prior to beginning the assignment to become familiar with the expectations for successful completion.

You are not required to submit this assignment to LopesWrite.

Hashing Quiz

Start Date & Time	Due Date & Time	Points	Time Limit
Sep 22, 2025, 12:00 AM	Sep 28, 2025, 11:59 PM	20	20 mins
			 30 mins

Assessment Traits

 Accommodated

 Timed

Assessment Description

Complete the quiz as directed by your instructor.

Week 4 Participation

Start Date & Time	Due Date & Time	Points
Sep 22, 2025, 12:00 AM	Sep 28, 2025, 11:59 PM	10

Topic 5: Spring Boot Security Configuration

Sep 29, 2025 - Oct 19, 2025 Max Points: 180

Security settings in Spring Boot, the popular web development framework, can be challenging to learn due to the complexity and variety of options available. In this lesson, you will learn how to transform an existing Spring Boot application into a secure application using Spring Security. This includes integrating necessary Spring Boot Security dependencies and configurations, creating a registration and login system, and restricting access to resources based on user authentication.

Objectives:

1. Implement user authentication and authorization mechanisms using Spring Security.
2. Utilize Thymeleaf Security Extras to conditionally render content based on user roles.
3. Implement password encoding using a secure hashing algorithm in the UserService class.
4. Identify key terminology and components used in Spring Boot security.
5. Debug security configurations to ensure proper access control and authentication.
6. Integrate Spring Boot Security dependencies and configurations in an application.
7. Restrict access to protected resources based on user authentication.

Resources

Debugging Security Issues (REQUIRED)

Refer to this resource when completing "Activity 5: Spring Security Configuration."

CST-407-RS-T5-Debugging-Security-Issues.docx

Activity 5: Spring Security Configuration (REQUIRED)

Refer to this resource when completing "Activity 5: Spring Security Configuration."

https://mygcuedu6961.sharepoint.com/:w:/r/sites/CSETGuides/CST/CST-407/CST-407-RS-T5-Activity-5-Spring-Security-Configuration.docx?d=wc39c0f36623e4aca8a22deace591233b&csf=1&web=1&e=ygg0Hx

Assessments

Activity 5: Spring Security Configuration

Start Date & Time	Due Date & Time	Points
Sep 29, 2025, 12:00 AM	Oct 5, 2025, 11:59 PM	40
	 Oct 19, 2025, 11:59 PM	

Assessment Traits

 Accommodated

Assessment Description

Refer to "Activity 5: Spring Security Configuration" and "Debugging Security Issues," located in the topic Resources, for details regarding successful assignment completion.

Access the file "CST-407-RS-T5-Activity-5-Spring-Security-Starting-Code.zip" as directed by the activity guide.

While APA style is not required for the body of this assignment, solid academic writing is expected, and documentation of sources should be presented using APA formatting guidelines, which can be found in the APA Style Guide, located in the Student Success Center.

This assignment uses a rubric. Review the rubric prior to beginning the assignment to become familiar with the expectations for successful completion.

You are not required to submit this assignment to LopesWrite.

Attachments

CST-407-RS-T5-Activity-5-Spring-Security-Starting-Code.zip

Wireshark Quiz

Start Date & Time	Due Date & Time	Points	Time Limit
Sep 29, 2025, 12:00 AM	Oct 5, 2025, 11:59 PM	20	20 mins
	 30 mins		

Assessment Traits

 Accommodated

 Timed

Assessment Description

Complete the quiz as directed by your instructor.

Week 5 Participation

Start Date & Time	Due Date & Time	Points
Sep 29, 2025, 12:00 AM	Oct 5, 2025, 11:59 PM	10

Spring Boot Security Quiz

Start Date & Time	Due Date & Time	Points	Time Limit
Oct 6, 2025, 12:00 AM	Oct 12, 2025, 11:59 PM	20	20 mins
			 30 mins

Assessment Traits

 Accommodated

 Timed

Assessment Description

Complete the quiz as directed by your instructor.

Week 6 Participation

Start Date & Time	Due Date & Time	Points
Oct 6, 2025, 12:00 AM	Oct 12, 2025, 11:59 PM	10

Milestone 1: Implement Spring Boot Security

Start Date & Time	Due Date & Time	Points
Sep 29, 2025, 12:00 AM	Oct 19, 2025, 11:59 PM	50
	 Nov 2, 2025, 11:59 PM	

Assessment Traits

 Accommodated

Assessment Description

Complete the milestone activity as directed by your instructor.

Refer to "CST-407 Milestone Assignments," located in Class Resources, for details regarding successful assignment completion.

While APA style is not required for the body of this assignment, solid academic writing is expected, and documentation of sources should be presented using APA formatting guidelines, which can be found in the APA Style Guide, located in the Student Success Center.

This assignment uses a rubric. Review the rubric prior to beginning the assignment to become familiar with the expectations for successful completion.

You are not required to submit this assignment to LopesWrite.

Attachments
CST-407-RS-T5-Milestone-Eventsapp.zip

CST-407 Review Quiz 1

Start Date & Time	Due Date & Time	Points	Time Limit
Oct 13, 2025, 12:00 AM	Oct 19, 2025, 11:59 PM	20	20 mins
			 30 mins

Assessment Traits
 Accommodated
 Timed

Assessment Description
Complete the quiz as directed by your instructor.

Week 7 Participation

Start Date & Time	Due Date & Time	Points
Oct 13, 2025, 12:00 AM	Oct 19, 2025, 11:59 PM	10

Topic 6: Implementing REST API Security with Java Web Tokens (JWT)

Oct 20, 2025 - Nov 9, 2025 Max Points: 180

This topic focuses on implementing REST API security using JSON Web Tokens (JWT) in a Java Spring Boot application. Starting from a REST API with no security, you will learn how to configure JWT for robust security. The primary focus is on generating and validating JWTs to secure API endpoints, ensuring that only authorized users can access protected resources. The topic also includes configuring Spring Security to leverage JWTs for both authentication and authorization processes. By the end of this topic, you will have a secured API service with JWTs, significantly enhancing the security of your RESTful APIs.

Objectives:

1. Implement JWT generation and validation for securing API endpoints.
2. Configure Spring Security to use JWTs for authentication and authorization.
3. Explain the structure and use of a JWT.
4. Analyze the structure and security practices of JSON Web Tokens.
5. Configure REST API endpoints to use JWT for secure communication.
6. Implement JWT Filters in Spring Security.
7. Identify potential vulnerabilities in web applications.

Resources

Activity 6: Implementing REST API Security JWT (REQUIRED)

Refer to this resource when completing "Activity 6: Implementing REST API Security JWT."

<https://mygcuedu6961.sharepoint.com/:w:/r/sites/CSETGuides/CST/CST-407/CST-407-RS-T6-Activity-6-Implementing-REST-API-Security-JWT.docx?d=wde45a9abcc14e9aa927e5152311c1e7&csf=1&web=1&e=S6ly81>

Assessments

JWT Quiz

Start Date & Time	Due Date & Time	Points	Time Limit
Oct 20, 2025, 12:00 AM	Oct 26, 2025, 11:59 PM	20	20 mins
			 30 mins

Assessment Traits

 Accommodated

 Timed

Assessment Description

Complete the quiz as directed by your instructor.

Week 8 Participation

Start Date & Time	Due Date & Time	Points
Oct 20, 2025, 12:00 AM	Oct 26, 2025, 11:59 PM	10

Activity 6: Implementing REST API Security JWT

Start Date & Time	Due Date & Time	Points
Oct 20, 2025, 12:00 AM	Nov 2, 2025, 11:59 PM	40
	 Nov 16, 2025, 11:59 PM	

Assessment Traits

 Accommodated

Assessment Description

Refer to "Activity 6: Implementing REST API Security JWT," located in the topic Resources, for details regarding successful assignment completion.

Access the files "CST-407-RS-T6.04 jquery api client.zip" and "CST-407-RS-T6.04ordersapp api starter.zip" as directed by the activity guide.

While APA style is not required for the body of this assignment, solid academic writing is expected, and documentation of sources should be presented using APA formatting guidelines, which can be found in the APA Style Guide, located in the Student Success Center.

This assignment uses a rubric. Review the rubric prior to beginning the assignment to become familiar with the expectations for successful completion.

You are not required to submit this assignment to LopesWrite.

Attachments

- CST-407-RS-T6.04 jquery api client.zip
- CST-407-RS-T6.04 ordersapp api starter.zip

CST-407 Review 2 Quiz

Start Date & Time	Due Date & Time	Points	Time Limit
Oct 27, 2025, 12:00 AM	Nov 2, 2025, 11:59 PM	20	20 mins
			 30 mins

Assessment Traits

-  Accommodated
-  Timed

Assessment Description

Complete the quiz as directed by your instructor.

Week 9 Participation

Start Date & Time	Due Date & Time	Points
Oct 27, 2025, 12:00 AM	Nov 2, 2025, 11:59 PM	10

Milestone 2: Implement JWT and REST API

Start Date & Time	Due Date & Time	Points
Oct 20, 2025, 12:00 AM	Nov 9, 2025, 11:59 PM	50
	 Nov 23, 2025, 11:59 PM	

Assessment Traits

-  Accommodated

Assessment Description

Complete the milestone activity as directed by your instructor.

Refer to "CST-407 Milestone Assignments," located in Class Resources, for details regarding successful assignment completion.

While APA style is not required for the body of this assignment, solid academic writing is expected, and documentation of sources should be presented using APA formatting guidelines, which can be found in the APA Style Guide, located in the Student Success Center.

This assignment uses a rubric. Review the rubric prior to beginning the assignment to become familiar with the expectations for successful completion.

You are not required to submit this assignment to LopesWrite.

XSS Quiz

Start Date & Time	Due Date & Time	Points	Time Limit
Nov 3, 2025, 12:00 AM	Nov 9, 2025, 11:59 PM	20	20 mins
			 30 mins

Assessment Traits

 Accommodated

 Timed

Assessment Description

Complete the quiz as directed by your instructor.

Week 10 Participation

Start Date & Time	Due Date & Time	Points
Nov 3, 2025, 12:00 AM	Nov 9, 2025, 11:59 PM	10

Topic 7: Cross Site Script Injection XSS

Nov 10, 2025 - Nov 23, 2025 Max Points: 150

This topic addresses the critical issue of cross-site scripting (XSS) vulnerabilities in web applications, which can compromise user data and application integrity. The focus is on understanding and demonstrating XSS attacks within a web application. Through hands-on activities, you will learn to identify and mitigate XSS vulnerabilities, thereby enhancing the security and reliability of web applications.

Objectives:

- 1. Demonstrate XSS attacks in a web application.
- 2. Differentiate between types of XSS attacks.
- 3. Debug vulnerable web applications for XSS issues.
- 4. Implement input validation and sanitization techniques to prevent XSS vulnerabilities.

Resources

Activity 7: Cross Site Script Injection (REQUIRED)

Refer to this resource when completing "Activity 7: Cross Site Script Injection."

<https://mygcuedu6961.sharepoint.com/:w:/r/sites/CSETGuides/CST/CST-407/CST-407-RS-T7-Activity-7-Cross-Site-Script-Injection.docx?d=wc5f8e0b7fb84424ca73af146023a5f22&csf=1&web=1&e=H9hg5U>

Assessments

Activity 7: Cross Site Script Injection

Start Date & Time	Due Date & Time	Points
Nov 10, 2025, 12:00 AM	Nov 16, 2025, 11:59 PM	40
	 Nov 30, 2025, 11:59 PM	

Assessment Traits

 Accommodated

Assessment Description

Refer to "Activity 7: Cross Site Script Injection," located in the topic Resources, for details regarding successful assignment completion.

Access the file "CST-407-RS-T7.comments.zip" as directed by the activity guide.

While APA style is not required for the body of this assignment, solid academic writing is expected, and documentation of sources should be presented using APA formatting guidelines, which can be found in the APA Style Guide, located in the Student Success Center.

This assignment uses a rubric. Review the rubric prior to beginning the assignment to become familiar with the expectations for successful completion.

You are not required to submit this assignment to LopesWrite.

Attachments

CST-407-RS-T7.comments.zip

CST-407 Review Quiz 3

Start Date & Time	Due Date & Time	Points	Time Limit
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Nov 10, 2025, 12:00 AM Nov 16, 2025, 11:59 PM 20 20 mins
 30 mins

Assessment Traits

 Accommodated

 Timed

Assessment Description

Complete the quiz as directed by your instructor.

Week 11 Participation

Start Date & Time	Due Date & Time	Points
Nov 10, 2025, 12:00 AM	Nov 16, 2025, 11:59 PM	10

Milestone 3: Implement Data Sanitization to Prevent XSS Attacks

Start Date & Time	Due Date & Time	Points
Nov 10, 2025, 12:00 AM	Nov 23, 2025, 11:59 PM	50
	 Dec 7, 2025, 11:59 PM	

Assessment Traits

 Accommodated

Assessment Description

Complete the milestone activity as directed by your instructor.

Refer to "CST-407 Milestone Assignments," located in Class Resources, for details regarding successful assignment completion.

While APA style is not required for the body of this assignment, solid academic writing is expected, and documentation of sources should be presented using APA formatting guidelines, which can be found in the APA Style Guide, located in the Student Success Center.

This assignment uses a rubric. Review the rubric prior to beginning the assignment to become familiar with the expectations for successful completion.

You are not required to submit this assignment to LopesWrite.

SQL Injection Quiz

Start Date & Time	Due Date & Time	Points	Time Limit
Nov 17, 2025, 12:00 AM	Nov 23, 2025, 11:59 PM	20	20 mins

Assessment Traits

 Accommodated

 Timed

Assessment Description

Complete the quiz as directed by your instructor.

Week 12 Participation

Start Date & Time	Due Date & Time	Points
Nov 17, 2025, 12:00 AM	Nov 23, 2025, 11:59 PM	10

Topic 8: SQL Injection

Nov 24, 2025 - Dec 14, 2025 Max Points: 235

In this topic, we will demonstrate SQL injection and how to prevent it. SQL injection is a common and serious security vulnerability that allows attackers to interfere with the queries that an application makes to its database. This can lead to unauthorized access to sensitive data, database manipulation, and other malicious activities.

Objectives:

1. Implement secure coding practices to prevent SQL injection attacks.
2. Compare the security implications of using JPA, Spring Data JDBC, and plain JDBC.
3. Identify what SQL injection is and why it is a significant security threat.
4. Identify parts of the code that are susceptible to SQL injection.
5. Identify potential XSS vulnerabilities in an application and demonstrate their impact.
6. Implement sanitization techniques to secure user input and prevent XSS attacks.
7. Conduct SQL injection attacks in a controlled environment to understand the potential impact.
8. Apply the principle of least privilege to database user accounts.
9. Explain why custom error pages improve security by avoiding exposing sensitive information through error messages.
10. Evaluate the security posture of a Christian organization.

Resources

Activity 8: SQL Injection and Prevention Java Spring Boot (REQUIRED)

Refer to this resource when completing "Activity 8: SQL Injection and Prevention Java Spring Boot."

<https://mygcuedu6961.sharepoint.com/:w:/r/sites/CSETGuides/CST/CST-407/CST-407-RS-T8-Activity-8-SQL-Injection-and-Prevention-Java-Spring-Boot.docx?d=wad53d371c25046a99ecf374d0a474185&csf=1&web=1&e=EJ9fDo>

Activity 9: Technology Security in a Christian Organization (REQUIRED)

Refer to this resource when completing "Activity 9: Technology Security in a Christian Organization."

Assessments

Benchmark - Activity 8: SQL Injection and Prevention Java Spring Boot

Start Date & Time	Due Date & Time	Points
Nov 24, 2025, 12:00 AM	Nov 30, 2025, 11:59 PM	40
	 Dec 14, 2025, 11:59 PM	

Assessment Traits

 Accommodated

 Benchmark

Assessment Description

Refer to "Activity 8: SQL Injection and Prevention Java Spring Boot," located in the topic Resources, for details regarding successful assignment completion. Doing this will help to minimize systems risks, threats, and vulnerabilities using best practices of secure programming.

While APA style is not required for the body of this assignment, solid academic writing is expected, and documentation of sources should be presented using APA formatting guidelines, which can be found in the APA Style Guide, located in the Student Success Center.

This assignment uses a rubric. Review the rubric prior to beginning the assignment to become familiar with the expectations for successful completion.

You are not required to submit this assignment to LopesWrite.

Benchmark Information *This benchmark assignment assesses the following programmatic competencies:*

BS in Software Development

1.4: Minimize systems risks, threats, and vulnerabilities using best practices of secure programming.

CST-407 Semester Review Quiz

Start Date & Time	Due Date & Time	Points	Time Limit
Nov 24, 2025, 12:00 AM	Nov 30, 2025, 11:59 PM	20	20 mins
			 30 mins

Assessment Traits

 Accommodated

Assessment Description

Complete the quiz as directed by your instructor.

Week 13 Participation

Start Date & Time	Due Date & Time	Points
Nov 24, 2025, 12:00 AM	Nov 30, 2025, 11:59 PM	10

Milestone 4: Demonstrate and Mitigate SQL Injection Vulnerability

Start Date & Time	Due Date & Time	Points
Nov 24, 2025, 12:00 AM	Dec 7, 2025, 11:59 PM	50
	 Dec 21, 2025, 11:59 PM	

Assessment Traits

 Accommodated

Assessment Description

Complete the milestone activity as directed by your instructor.

Refer to "CST-407 Milestone Assignments," located in Class Resources, for details regarding successful assignment completion.

While APA style is not required for the body of this assignment, solid academic writing is expected, and documentation of sources should be presented using APA formatting guidelines, which can be found in the APA Style Guide, located in the Student Success Center.

This assignment uses a rubric. Review the rubric prior to beginning the assignment to become familiar with the expectations for successful completion.

You are not required to submit this assignment to LopesWrite.

Week 14 Participation

Start Date & Time	Due Date & Time	Points
Dec 1, 2025, 12:00 AM	Dec 7, 2025, 11:59 PM	10

Activity 9: Technology Security in a Christian Organization

Start Date & Time	Due Date & Time	Points
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Nov 24, 2025, 12:00 AM Dec 14, 2025, 11:59 PM 40

 Dec 28, 2025, 11:59 PM

Assessment Traits

 Accommodated

Assessment Description

Refer to "Activity 9: Technology Security in a Christian Organization," located in the topic Resources, for details regarding successful assignment completion.

While APA style is not required for the body of this assignment, solid academic writing is expected, and documentation of sources should be presented using APA formatting guidelines, which can be found in the APA Style Guide, located in the Student Success Center.

This assignment uses a rubric. Review the rubric prior to beginning the assignment to become familiar with the expectations for successful completion.

You are not required to submit this assignment to LopesWrite.

Milestone 5: Implement Network Security with SSL

Start Date & Time	Due Date & Time	Points
Nov 24, 2025, 12:00 AM	Dec 14, 2025, 11:59 PM	55
	 Dec 28, 2025, 11:59 PM	

Assessment Traits

 Accommodated

Assessment Description

Complete the milestone activity as directed by your instructor.

Refer to "CST-407 Milestone Assignments," located in Class Resources, for details regarding successful assignment completion.

While APA style is not required for the body of this assignment, solid academic writing is expected, and documentation of sources should be presented using APA formatting guidelines, which can be found in the APA Style Guide, located in the Student Success Center.

This assignment uses a rubric. Review the rubric prior to beginning the assignment to become familiar with the expectations for successful completion.

You are not required to submit this assignment to LopesWrite.

Week 15 Participation

Start Date & Time	Due Date & Time	Points
Dec 8, 2025, 12:00 AM	Dec 14, 2025, 11:59 PM	10